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M 203 AIR REMOVAL

1. SCOPE OF WORK

The Bidder shall supply the following for one block in accordance with **Part 4, Section 1 Subsection 1.2 "Scope of work"**.

- Two (2) 100% duty main steam jet air ejectors for condenser steam side service complete with all appurtenance
- One (1) starting Air Ejector for condenser complete with all appurtenances.
- One (1) C.W. Priming System for the water boxes in condenser complete with all appurtenances including vacuum pump and motor (if required).

The air ejector shall include, but not be limited to the following:

- (1) Steam Jet Air Ejector
 - (a) Ejector with coolers
 - (b) Manometer
- (2) Starting Air Ejector
 - (a) Ejector
 - (b) Silencer
- (3) Foundation Bolts
- (4) Drain and Vent Piping, valves and traps
- (5) Instruments
- (6) Insulation
- (7) Painting
- (8) Special Tools and Standard Tool Sets
- (9) Lifting Lug

Instead of air ejectors water ring pumps can be offered for steam side evacuation with a scope of work, comfort and quality equivalent to the specified air ejectors and to the approval of the Owner.

C.W. Priming pump shall include, but not be limited to, the following:

- (1) Vacuum Pump

- (2) Electric Motor
- (3) Coupling and Guard
- (4) Water Tank
- (5) Sealing system
- (6) Bedplate
- (7) Instrument and Control Equipment
- (8) Electrical Equipment

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 “Codes and Standards”**. The codes and standards of the following organizations extracted from **Part 4, Subsection 2.5 “Codes and Standards”** or international equivalent standards are specifically applicable to the design manufacturer and testing of the equipment included in this Specification.

ASME	American Society of Mechanical Engineers.
ANSI	American National Standards Institute.
ASTM	American Society of Testing and Materials.
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HEI	Heat Exchange Institute.
IEC	International Electrotechnical Commission.
JIS	Japan Industrial Standard
NEMA	National Electrical Manufacturers Association

3. DESIGN AND OPERATING CONDITIONS

Two (2) x 100% duty two stage Steam air ejectors shall be installed for condenser to eject the leaked or deaerated air in the turbine cycle. One air ejector will operate as a base unit and the other air ejector will be on stand by.

The inter and after condensers shall be installed with the steam jet air ejector, and the ejector steam shall be condensed by the condensate, and returned to

the surface condenser.

One (1) starting air ejector shall be installed to build up the condenser vacuum before turbine starting.

The ejector steam shall be supplied from the high pressure auxiliary steam source in case of auxiliary steam system alternatively the HRSG shall be able to cater to motive steam requirement of ejectors from HP steam header, so that condenser vacuum can be established before bypasses are taken into operation.

The design shall be in accordance with the “Standard for Steam Surface Condenser of the Heat Exchanger Institute” and this specification.

The C.W. priming pump (if necessary) shall be designed to evacuate air from the top part of condenser and closed cooling water heat exchanger water boxes to establish the siphon effect over the condenser, etc. within 30 minutes. If necessary, continuous air removal equipment as per requirement of **Part 4, Section 3, M-202 Clause 4.3** shall be provided.

4. DETAILED REQUIREMENTS

4.1 Main Steam Jet Air Ejector

Each air ejector shall operate efficiently when supplied with pressure reduced steam the source of steam being the H.P. steam auxiliary manifold.

The steam supply for the main ejectors shall include a common motorized isolating valve with provision for manual operation and pressure regulating device(s) with associated bypass pipe(s) and relief valve discharging to the main exhaust system. Each air ejector shall in addition be supplied with a motorized isolating valve in its common steam supply pipe and also a separate isolating valve and strainer for each ejector stage.

The air ejectors shall be provided with surface type condensers connected in to the main condensate circuit.

All necessary air extraction piping between the main condenser connection, and the air ejectors and the air outlet piping including the main atmospheric exhaust pipe shall be within this contract, together with all necessary isolating and non- return valves in the outlet pipes from each ejector.

Suitable drains for the ejectors and piping system shall be provided to return the drains to the condenser and clean drains tank or alternatively reject the drains to waste. The drains to waste shall be suitable led to the nearest drainage trench. Adequate drains shall be fitted to allow rapid changeover from one ejector unit to the other. The drains system to be provided shall be complete including all necessary drain traps, strainers, loop seals and gauge glasses.

The air ejector shall be conveniently arranged for maintenance and operation with control valves and other related equipment easily accessible from platforms or floors.

A gauge panel associated with each main ejector shall be provided and shall include all appropriate local instrumentation. The panel shall be mounted in a convenient position to the approval of the Owner.

4.2 Starting Air Ejector

A suitable single stage, steam jet, starting air ejector shall be provided capable of reducing the condenser pressure to at least 340 mbar (abs) within 10 minutes of being put into operation when supplied with steam at pressures as specified for the main air ejectors, and the turbine shaft gland sealing system being operated in accordance with the agreed start-up procedure.

The starting air ejector shall be of sufficient capacity to ensure adequate removal of air from the condenser shells, the casings of the turbine and all pipe-works normally connected there too.

All necessary air extraction and exhaust pipework shall be included together with isolating and non-return valves, as specified for the main air ejectors.

A gauge panel shall be provided and shall include all appropriate local instrumentation. The panel shall be mounted in convenient position to the approval of the Owner.

Exhaust piping of starting air ejector shall be installed up to 3 m above the main building roof.

Noise at exit opening of starting air ejector exhaust pipe shall be in accordance with **Part 4, Section 2.9**.

4.3 C.W. Priming Pump

All parts of the vacuum pumps shall be of a design and construction which has proven its durability, reliability, and continuous efficiency in units of similar type under similar service.

The equipment must be designed to operate over extended periods of time without maintenance beyond normal bearing oiling, and packing adjustments.

All equipment furnished under this Specification shall be of type, material and design suitable for the operating conditions and services intended and shall meet with the approval of the Owner.

4.3.1 Pump

The vacuum pump shall be of the rotary, water sealed type.

The vacuum pump shall be provided with a drain trap for suction line, inlet check valve, strainer, solenoid shut off valve and isolation valves.

The U-seal piping shall be installed at connection piping between the condenser water boxes and the vacuum pump to shut off the sea water to vacuum pump.

The vacuum pump shall be operated locally.

The vacuum pump shall be of the completely self-contained package type insofar as possible, furnished complete with all integral automatic controls. The vacuum pump and associated equipment, shall be furnished mounted on a base complete with motor drive and shall be equipped with dynamically balanced flexible coupling.

Bidder shall be responsible for determining that the pump and its motor drive as a unit operate in dynamic balance without vibration.

Sufficient WR^2 shall be incorporated in the vacuum pumps to limit motor current pulsation, at all loads, to the value recommended by IEEE.

4.3.2 Water Tank

The water tank shall be installed to store the seal water of vacuum pump, and shall be provided with a float valve to control tank water level.

The tank shall be provided with a glass gauge, solenoid shut off valves, drain valves, drain piping and all necessary connections.

The material of tank shall be of anticorrosive type such as rubber lining.

4.4 Materials

Material shall be equal or superior quality to the material specified bellow.

4.4.1 Steam Jet Air Ejector

- (a). Suction chambers Carbon steel
- (b). Steam nozzle Stainless steel
- (c). Diffuser Carbon steel
- (d). Shell and Water box of condenser Carbon steel
- (e). Tube sheet Carbon steel
- (f). Tubes Stainless steel

4.4.2 Starting Air Ejector

- (a). Suction chamber..... Carbon steel
- (b). Steam nozzle..... Stainless steel

4.4.3 C.W. Priming Pump

The main part materials shall be as follows, and other parts materials shall be suitable for use with see water vapor.

- (a). Casing..... Cast iron
- (b). Rotor Bronze cast
- (c). Shaft Stainless steel

4.5 Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter, and shall be given one prime coat of paint, suitable for the service and environment. The color of the prime coat be the Bidder's standard. No paint having an asphaltum base shall be used.

The Bidder shall ship a sufficient quantity of paint for touching up where prime coat has been damaged during shipment and erection. Paint shall be the same kind and quality as used each piece of equipment in the shop.

4.6 Insulation

Refer to Part 4 Section 3, M-003 “Thermal Insulation and Logging”

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Shop Test

Each pump shall be stop tested by the Bidder as follows:

- Free dry air capacity of each pump shall be measured at a minimum of five test points at full design speed and temperature.
- Each pump shall be subjected to a hydrostatic test at 1.5 times the design pressure and full vacuum.

5.2 Site Test

After equipment is put in service, the Bidder shall run a test to ascertain that the equipment is performing according to the guarantee in every respect. The test shall be run in accordance with the ASME Test Code for the Displacement Compressors, Vacuum Pumps and Blowers. Any deficiencies brought out by this test shall be promptly corrected by Bidder at no cost to Owner.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality control and documentation shall conform to the requirements of **Part 4, Subsection 2.7**.

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the design, supply and installation of complete instrumentation and control system and devices for the Condenser Vacuum and CW Priming including but not limited to those listed and described in this Technical Specification.

The Bidder shall also furnish and install all necessary devices in the control systems for the stopping and starting of the Condenser Vacuum Pumps priming and/or continuous air removal equipment for the CW side (including status indication) from the main control desk.

7.2 Local Control Panel

Refer to **Part 4, Section 3, I-002, "Local Control Panels"** for the design requirements for the local control panels.

8. ELECTRICAL SPECIFICATION

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Condenser Vacuum Pump.

For general electrical requirements, see **Part 4, Section 3, E-001**.

8.2 Motors

For detailed electrical requirements, see **Part 4, Section 3, E-210** for Low Voltage Motor.

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